DATA STRUCTURE

LAB PROGRAMS

10. a) Stack – Push operation using array

```
#include <stdio.h>
#define MAX 5
int main()
{
    int stack[MAX] = {10, 20, 30};
    int top = 2;
    int value, i;
    printf("Enter value to push: ");
    scanf("%d", &value);
    if(top == MAX - 1)
    {
        printf("Insertion is not possible. Stack Overflow");
    }
    else
    {
        top++;
        stack[top] = value;
        printf("Stack after Push:\n");
        for(i = top; i >= 0; i--)
        {
            printf("%d\n", stack[i]);
        }
    }
```

```
    return 0;
}
```

```
Output
Enter value to push: 45
Stack after Push:
45
30
20
10


=== Code Execution Successful ===
```

b) Stack – Pop operation using array

```
#include <stdio.h>
#define MAX 5
int main()
{
    int stack[MAX] = {10, 20, 30};
    int top = 2;
    int i;
    if(top == -1)
    {
        printf("Deletion is not possible. Stack Underflow");
    }
    else
```

```
    {
        printf("Deleted Element = %d\n", stack[top]);
        top--;
        printf("Stack after Pop:\n");
        for(i = top; i >= 0; i--)
        {
            printf("%d\n", stack[i]);
        }
    }
    return 0;
}
```

Output

```
Deleted Element = 30
Stack after Pop:
20
10


=== Code Execution Successful ===
```

c) Stack – Display operation using array

```
#include <stdio.h>
#define MAX 5
int main()
{
```

```c
    int stack[MAX] = {10, 20, 30};
    int top = 2;
    int i;
    if(top == -1)
    {
        printf("Stack is Empty");
    }
    else
    {
        printf("Stack Elements are:\n");
        for(i = top; i >= 0; i--)
        {
            printf("%d\n", stack[i]);
        }
    }
    return 0;
}
```

Output

```
Stack Elements are:
30
20
10


=== Code Execution Successful ===
```